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Report Title: SEG3125 Assignment 5 — Analytics Page Report

Goal + Data

a. Domain and Dataset

My dashboard is in the sports and fitness domain, with a focus on powerlifting performance and nutrition. The strength-comparison section uses the official Canada Powerlifting National Records (<https://www.powerlifting.ca/>) database. I selected the Classic records for men and women across different age divisions and bodyweight classes. For the dashboard, I retained only the squat, bench press, and deadlift values. The selected values were manually organized into a JavaScript object so that the chart can update when the user changes the gender category, age division, or bodyweight class. Because national records may change, the dashboard identifies the data as a static snapshot and provides a link to the official source.

The nutrition section uses calculated data rather than a downloaded dataset. Basal metabolic rate is estimated using the Mifflin–St Jeor equation (<https://pubmed.ncbi.nlm.nih.gov/2305711/>), based on the user's gender category, age, height, and weight. The estimated maintenance calories are then calculated using the activity factor selected by the user. The default macronutrient distribution is 50% carbohydrates, 20% protein, and 30% fat, which falls within Health Canada's adult macronutrient reference ranges (https://www.canada.ca/en/health-canada/services/food-nutrition/healthy-eating/dietary-reference-intakes/tables/reference-values-macronutrients.html?utm_source=chatgpt.com).

No synthetic or AI-generated dataset is presented in the dashboard. Generative AI was used to support coding, translation, and data organization, but the powerlifting records and nutrition references came from the sources identified above.

b. Inspiration Dashboard

The main inspiration for my dashboard was the provided official MLS Player Stats page (<https://www.mlssoccer.com/stats/players/#season=MLS-SEA-0001KA&competition=mls-regular-season&club=all&statType=general&position=all>). I was inspired by its clear sub-navigation, filter controls, structured statistical table, and strong visual hierarchy. In particular, the page highlights selected information while keeping a large amount of data organized and easy to scan. I adapted this approach to the powerlifting domain by using two sub-navigation sections, structured input controls, result summaries, interactive charts, and detailed comparison tables.

Reflection/Design

a. Charts

The dashboard includes two different types of interactive charts. The first chart is a line chart that compares the user's squat, bench press, and deadlift results with Canadian Classic national records. The horizontal axis represents the three competition lifts, while the vertical

axis represents the weight lifted in kilograms. I selected a line chart because it clearly shows the user's overall strength profile across the three lifts and makes the difference between personal results and national records easy to identify.

The second chart is a 100% horizontal stacked bar chart that presents the user's macronutrient distribution. It displays the percentages of daily calories assigned to carbohydrates, protein, and fat. I selected this chart type because all three nutrients are parts of one total daily calorie target, and the stacked format makes their proportions easy to compare.

b. 3Cs

For Context, each chart includes a clear title, a short explanation, axis labels, units, legends, and supporting information. The strength chart identifies that the values are Canadian Classic individual lift records and shows the selected gender category, age division, and bodyweight class. The nutrition section explains that the results are estimates and shows the user's basal metabolic rate, maintenance calories, selected activity factor, and macronutrient calculations.

For Clutter-Free design, I limited each chart to the information needed for the main comparison. The strength chart contains only two data lines and three lift categories. Unnecessary decorative elements and excessive grid lines were avoided. The nutrition chart uses one horizontal bar divided into three sections, which keeps the visualization simple. Detailed numerical information is placed in tables below the charts instead of overcrowding the charts.

For contrast, I used the website's existing black, red, and white colour palette. Red is used to emphasize Canadian records, important results, active navigation items, and selected controls. Black and red make the two datasets in the strength chart visually distinct. In the nutrition chart, separate colours identify carbohydrates, protein, and fat. Large headings, result cards, bold labels, and white space also create a clear visual hierarchy.

c. Layout, Title, and Interactions

The page title, Athlete Stats, clearly reflects the dashboard's focus on both powerlifting performance and nutrition. A sub-navigation bar divides the website into two main sections: Strength Comparison and Nutrition & Calories. This prevents the page from becoming overcrowded and allows users to focus on one task at a time. Both sections follow a consistent structure with an input form, summary cards, an interactive chart, and a supporting data table.

In the strength section, users can modify the gender category, age division, bodyweight class, and their squat, bench press, and deadlift results. The available bodyweight classes automatically change based on the selected gender and age division. After selecting Generate Comparison, the line chart and detailed table update to compare the user's results with the corresponding Canadian Classic national records.

In the nutrition section, users enter their gender, age, height, weight, and activity level to calculate their estimated BMR and maintenance calories. They can also adjust the percentages of carbohydrates, protein, and fat. The system checks that the three percentages equal 100% before updating the stacked bar chart and nutrition table. Hover tooltips, reset buttons, and input validation provide immediate feedback and help users correct or modify their selections easily.

d. Internationalization

The dashboard supports English and French. English was used as the original language, while the French translations were created with AI assistance and then reviewed to keep powerlifting, nutrition, chart, and interface terminology consistent. The language selector updates the navigation, form labels, buttons, chart titles, legends, tooltips, tables, and explanatory text.

One difficulty was that French text is often longer than English. To prevent layout problems, I used flexible containers, responsive spacing, and wider controls instead of fixed text widths.

The localization process also included Canadian number formatting through `Intl.NumberFormat`, metric units such as kilograms and centimetres, and translated age divisions and activity descriptions. I used text-based language buttons rather than national flags because one language can be used in multiple countries.

Link to UI

ziye-seg3125-project.netlify.app

Link to GitHub Repository

https://github.com/Ziye-Zhou/SEG3125_Project

High-Fidelity Prototype visual design choices

The dashboard uses the same black, red, and white visual style as the rest of the MN Powerlifting portfolio. Black creates a strong background, red highlights active controls and important results, and white improves readability and contrast. The Font Michroma is used for major headings to create a modern and technical appearance, while Source Sans Pro is used for navigation, forms, tables, and descriptions because it is clear and readable. The Font Squada One is used selectively for shorter labels and card titles to reinforce the strength-training theme.

The interface uses consistent input panels, result cards, buttons, charts, and tables across both dashboard sections. Clear spacing and visual hierarchy help users move from entering information to reviewing the generated results. The design is also responsive, so navigation, forms, charts, and tables adjust for smaller screens. This consistency makes the Analytics Site feel connected to the Service Site, Memory Game, E-commerce Site, and the overall portfolio.

I used AI to support both the high-fidelity prototype and the report. For the prototype, AI helped me plan the unsure dashboard structure, generate and revise React, HTML, and CSS code, create the English–French interface, and troubleshoot navigation, dropdown and chart issues. It also helped me locate the Canadian powerlifting record data and nutrition reference sources, then organize the selected squat, bench press, deadlift, calorie, and macronutrient information for use in the dashboard. I reviewed the sources and adjusted the data and code to match the final design.

For the report, I used AI to organize the required sections, improve clarity, shorten or expand paragraphs, and describe the charts, and internationalization. I reviewed and edited the generated content so that it accurately reflects my final dashboard and design decisions.